

AI / BACKEND ENGINEER

ALI HASSANE YAYA

Berlin, Germany

mail yayasafihas@gmail.com

tel +49 152 57083662

in/ [ali-hassane-yaya](#)

gh/ [yaya-ali](#)

web yaya-ali.github.io

PROFESSIONAL SUMMARY

Backend-focused AI / software engineer specializing in Python and FastAPI, with a year building a production OpenAI GPT-4 translation system for an EdTech platform, full-stack delivery of a university flagship banking web/app, and earlier Docker/production work. Now deepening LLM engineering by building projects in the open — cost-aware model routing, multi-agent workflows (LangChain / LangGraph), RAG and guardrail testing. Strong across the delivery lifecycle (API design, data modeling, CI/CD); MSc Data Science student in Berlin with a competitive security/CTF background (1st place, IIUM CTF).

PROFESSIONAL EXPERIENCE

Full-Stack Developer — XtraLesson (EdTech)

Mar 2025 – Feb 2026 · Qatar · Remote

Python · FastAPI · OpenAI GPT-4 · PostgreSQL · async pipelines · CI/CD

- Built an async, fault-tolerant OpenAI GPT-4 translation system for a 50-course library, with retry logic that reprocessed only failed blocks — cutting API calls by roughly a third and turning days of processing into hours.
- Implemented a real-time parsing/job tracker (Start → Process → Complete) that gave non-technical teams live visibility of translation progress and removed manual status requests.
- Refactored translation into user-triggered actions and customized frontend branding to XtraLesson's identity, improving flexibility and engagement.

App & Web Developer — IIUM Flagship 18: Sustainable Banking

2023 · ~6 months · International Islamic University Malaysia · Kuala Lumpur

Full-stack web & mobile · deployment · domain migration

- Sole developer for the university's Flagship 18 sustainable-banking initiative — built the entire website and mobile application end to end.
- Migrated the platform to its own dedicated domain, handling deployment and the full transition.

Junior Software Engineer — WenakLabs

Mar 2020 – Feb 2021 · N'Djamena, Chad · On-site

Docker · production systems · release & infrastructure

- Developed and maintained applications supporting core production systems.
- Managed containerization with Docker to streamline deployment and scalability.
- Collaborated on infrastructure and release processes to ensure system reliability.

EDUCATION

MSc Data Science & Business Intelligence

GISMA Business School, Berlin, Germany · GPA 3.5

Oct 2025 – present

BSc Information Technology (Hons.) — Business Intelligence & Information Science

International Islamic University Malaysia (IIUM) · GPA 4.0 / CGPA 3.2

2021 – 2025

PROJECTS (BUILDING IN THE OPEN)

RouteIQ — LLM Cost Router & Observability Gateway **SHIPPED** · Python, FastAPI, Redis, Prometheus, Grafana · github.com/yaya-ali/routeiq

- Shipped an API gateway that routes each LLM request to the cheapest capable model — measured **10.2% cost saving with 0% quality drop** on an eval set, enforced by an eval-gated CI pipeline that blocks quality regressions.
- Per-team daily budgets via atomic date-keyed Redis counters (HTTP 429 on exhaustion, UTC-midnight reset); Prometheus metrics with auto-provisioned Grafana dashboard; chat and quiz demo UIs as consumer apps.

SentinelAI — AI Red-Team & Guardrail Harness **IN PROGRESS** · Python, GitHub Actions

- Building a layered guardrail / red-team harness (regex, rule-based and model-based defenses) with a CI suite that catches regressions; measuring jailbreak-rate reduction.

ClinicalMind — Multi-Agent Clinical Triage API **IN PROGRESS** · FastAPI, LangGraph, LlamaIndex, pgvector

- Building a multi-agent triage service with structured outputs, hybrid retrieval and evaluation gates.

TECHNICAL SKILLS

Languages	Python, SQL, TypeScript, JavaScript, PHP
Backend & APIs	FastAPI, Pydantic, REST API design, microservices, asyncio & async job queues, Node.js / Express familiar with Flask
AI / LLM	LLM application engineering, RAG, embeddings & vector search (pgvector), prompt engineering, multi-agent systems (LangChain, LangGraph, LlamaIndex), evaluation harnesses, guardrails / red-teaming, model routing, OpenAI & Anthropic APIs, Hugging Face, PyTorch, scikit-learn
Databases	PostgreSQL, Supabase (pgvector), Redis, MongoDB — schema design, query optimization, indexing
Cloud & DevOps	AWS, Azure, Google Cloud, Docker, Kubernetes, GitHub Actions (CI/CD), Terraform, Prometheus, Grafana
Frontend	React, Next.js, Tailwind CSS
Practices	Agile / Scrum, code review, automated testing (pytest), Git, Postman, Linux

CERTIFICATIONS & LANGUAGES

Certifications: Cisco CCNA · AWS Cloud Practitioner · FastAPI & REST APIs (Udemy) · Intro to Cybersecurity (Cisco) · 1st Place — IIUM CTF (2022) · Top 20% — PicoCTF Global (2023)

Languages: Arabic (native) · English (fluent) · French (intermediate) · German (basic, improving)